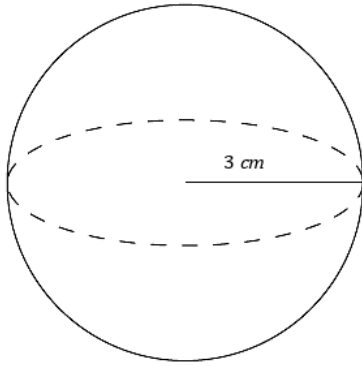


Volume of spheres and hemispheres

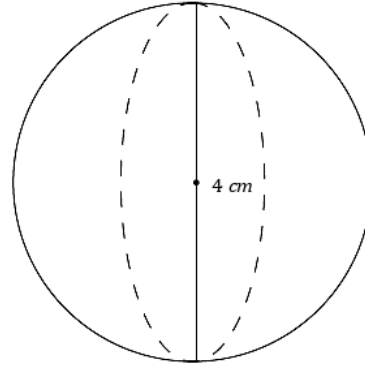


1 Find the volume of the following spheres, correct to two decimal places:

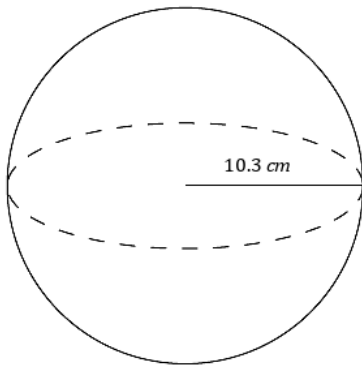
a



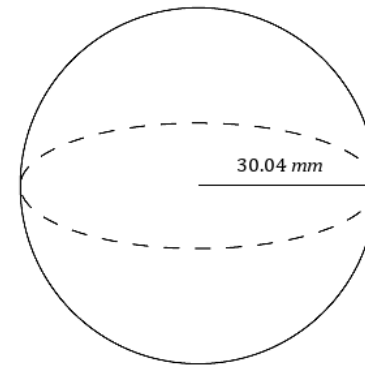
b



c



d



2 Find the radius of the following spheres, correct to two decimal places:

a A sphere with a volume of 167 cm^3 .

b A sphere with a volume of $\frac{343\pi}{3} \text{ cm}^3$.

3 The planet Mars has a radius of 3400 km.

Find the volume of Mars. Write your answer in scientific notation to three decimal places.

4 The planet Jupiter has a radius of 69 911 km, and planet Mercury has a radius of 2440 km. Assume that both planets are spheres.

How many times bigger is the volume of Jupiter than Mercury? Round your answer to one decimal place.

5 A ball has a volume of 904.779 cubic units.

Find the radius of the ball. Give your answer correct to the nearest tenth.

- 6 The radius of a bowling ball is 10.9 cm.



Find the volume of the ball. Round your answer to three decimal places.

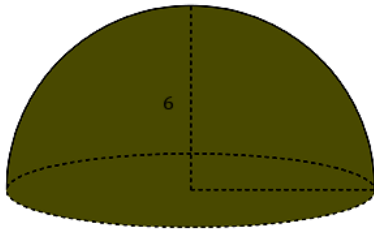
- 7 The volume of a sphere is $20\,579.526\text{ cm}^3$.

Find the diameter of the sphere. Give your answer correct to one decimal place.

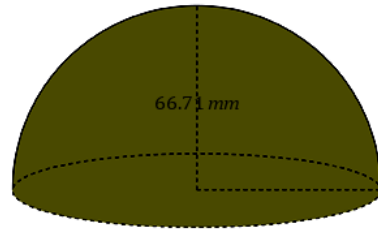
- 8 If the radius of a sphere doubles, by what factor will its volume increase?

- 9 Find the volume of the following solids, correct to two decimal places:

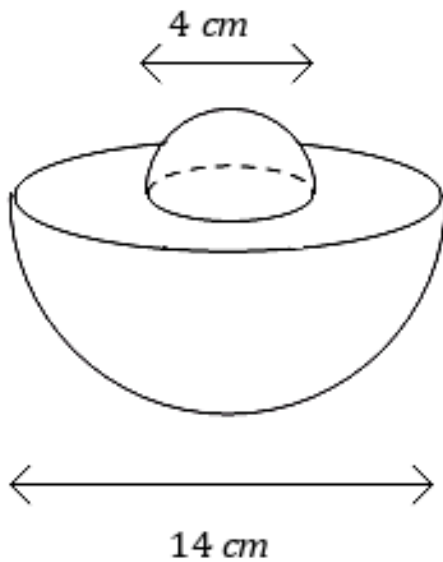
a



b



c

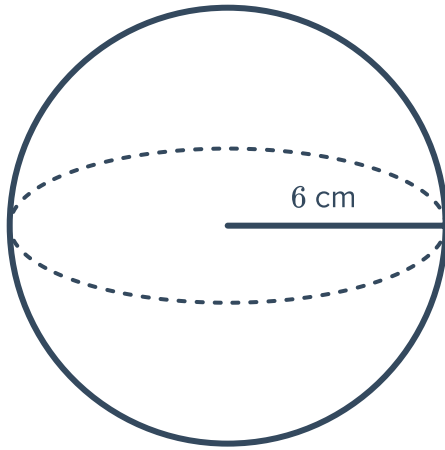




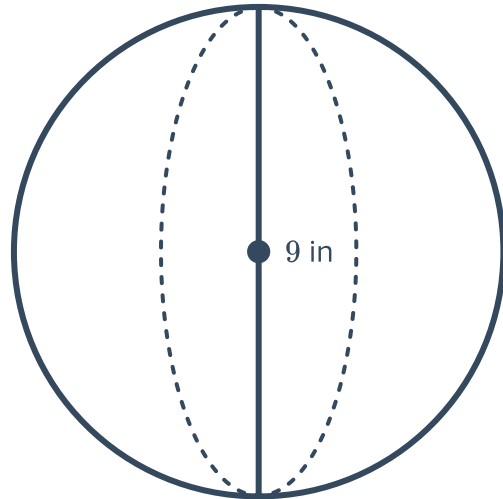
Additional questions

10 For each sphere, calculate the volume:

a



b



11 The diameter of a bowling ball is 8.6 in. What is the approximate volume of the ball?

12 The volume of a sphere is $972\pi \text{ yd}^3$. What is the diameter of the sphere?

13 Tennis balls are packaged and sold in cylindrical containers, with three balls per container. The height of a standard tennis ball container is 21 cm.

- a What is the volume of a tennis ball?
- b What is the volume of the container?
- c What percent of the container is occupied by the tennis balls?